

VAT Identifier Discovery

I started the project by understanding the task and identifying the steps needed to solve it.

The client has 40,000 UK suppliers. For some of these companies, the VAT registration number is already known, while for the remaining companies it needs to be discovered and checked to be correct.

Firstly, I identified a smaller dataset that could provide a representative set of UK companies to work with which has been provided by Companies House. Instead of using the entire set I decided to randomly extract only 200 companies.

The next step was to try and find VAT's for these companies by web searching. I started with the first 10 companies from the data for which I manually searched the VAT. In all cases, Google's AI search results were the most useful starting point, as they provided potential VAT numbers together with links to the webpages where the information was found. I also tried to look to other sources, as not to base the result only on the AI because it can be wrong. I tried the filling history pdf's provided by Companies House, but this turned out to be a dead end as it was very time consuming and in all the cases I could not find VAT's in there.

After finding a possible VAT the next step was to verify it using the HMRC checker. The HMRC told me whether the VAT was valid and to which company it belonged. After this a very important step was to check if the company provided by HMRC was the one I was actually searching for and I did this by comparing the companies number and address. This was very important to do as there have been a case where I found 2 companies with the exact same name and one had a public VAT and the other didn't.

After completing the manual searched I documented the results in a table in order to keep the discovered VAT numbers, their sources and their HMRC check results in a structured format. I found that out of the first 10 test cases, I found 1 positive result, 2 false-positives and 7 negative results. This means that, I successfully identified and validated a VAT number for 1 out of 10 companies (10%). The results also demonstrate an important limitation of the discovery process as VAT numbers found online cannot automatically be assigned to a company simply because the name appears similar. It must first pass HMRC validation and then be checked against the identity of the target company. The manual search also showed that publicly available VAT information is inconsistent, as some companies disclose their VAT number online, while others don't.

After research, the next step would be to determine how the process could be scaled from 10 companies to 200 and then to the full dataset of 40,000 companies. Based on these results, the process could be automated by developing a program capable of searching the web for VAT numbers for the remaining companies and, where possible, submitting the discovered candidates for HMRC validation. However, I believe the submitted VAT's must be manually checked, as I found cases where a VAT was associated to complete other companies that had the same name as my target. Once the process has been tested and refined on the sample, it could be scaled to the full dataset of 40,000 companies.

The companies for which a VAT number is successfully discovered and validated could then be included in a completed dataset. For the companies for which no VAT number can be publicly discovered, the process becomes harder. One possible solution would be to contact these companies directly and request their VAT registration number, but I would expect this approach to have a low response rate and would therefore require significant resources. Another possible approach would be to identify a service or database through which additional company information could be requested. This could potentially provide information that is not publicly available, but it would require approval of the request and would introduce additional time and cost to the process. For this project, I focused on the companies for which a VAT number could be found on the web.

If additional resources were available, I would consider implementing the approaches mentioned earlier in order to assure a more complete dataset. In a larger team, the remaining companies could also be divided among team members to make the process more efficient. For example, if half of the 40K companies remained without a discovered VAT number, the remaining dataset could be divided into smaller portions and assigned to pairs of team members. One person could focus on contacting companies directly, while the other could investigate external databases or submit information requests. These activities could possibly also be automated to reduce the amount of manual work required. But again, the results obtained would have to be checked.